

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Application Based Question

Course Code:	CSA5110	Course Title:	Cryptography and Network Security
CO Assessed:	CO1-Imitation (BL3)	Assessment:	Assessment Tool 1– Application Based Question
Weightage:	40% of CIA	Total Marks:	25
CO1 Statement:	<i>Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)</i>		
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Section / Batch:		Date:	16-07-2026

Code of Conduct

I M. Mokshith Reddy - 192372373 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- **This assessment contains 5 questions.**
- **Each question carries 10 marks. Total: 50 Marks.**
- **Ensure clarity, structure, and logical flow in answer**
- **Duration: 50 Minutes.**

Question s	Understandi ng of Concepts(25)	Experimenta l design(30)	Use of Tools(20)	Analysi s of Result (15)	Documentatio n (10)	Total Mark s
Overall Total (100)						

Question 1: Security Attacks

Problem Statement

A multinational financial institution experienced several security incidents in its online banking system. The task is to identify the type of attack, classify it as passive or active, analyze its impact on the CIA triad, and recommend preventive measures.

Incident Analysis

- **Incident 1: Fake Emails Requesting Login Credentials**

Attack Name: Phishing Attack .

Attack Type: Active Attack

- **Description**

Phishing is a social engineering attack where attackers pretend to be legitimate organizations to steal user credentials through fake emails or websites.

- **Impact on CIA**

Confidentiality

Customer usernames and passwords are stolen.

Integrity

Attackers can change banking information after gaining access.

Availability

Customers may lose access to their accounts if passwords are changed.

- **Preventive Measures**

- Multi-Factor Authentication (MFA)
- Anti-phishing email filters
- Security awareness training
- Spam filtering
- Secure email gateways
-

- **Incident 2: Unauthorized Fund Transfers Using Session Tokens**

Attack Name: Session Hijacking

Attack Type: Active Attack

- **Description**

Attackers steal valid session tokens and impersonate authenticated users to carry out unauthorized banking transactions.

- **Impact on CIA**

Confidentiality

User session information is exposed.

Integrity

Unauthorized fund transfers change banking records.

Availability

Genuine users may temporarily lose access to their accounts.

- **Preventive Measures**

- HTTPS/TLS encryption
- Secure cookies (HttpOnly and Secure)
- Session timeout
- Token regeneration after login
- VPN for secure communication

- **Incident 3: Repeated Traffic Spikes on Banking Server**

Attack Name: Distributed Denial-of-Service (DDoS)

Attack Type: Active Attack

- **Description**

Attackers overwhelm the server with excessive traffic, causing banking services to be unavailable.

- **Impact on CIA**

Confidentiality

No direct impact.

Integrity

Banking records stay the same.

Availability

Banking services become unavailable.

- **Preventive Measures**

- Firewall
- Intrusion Prevention System (IPS)
- Load Balancer
- DDoS Protection Services
- Traffic Rate Limiting

- **Incident 4: Weak Employee Passwords**

Attack Name: Password Attack (Brute Force/Dictionary Attack)

Attack Type: Active Attack

- **Description**

Attackers guess passwords using automated tools.

- **Impact on CIA**

Confidentiality

Sensitive employee information is exposed.

Integrity

Attackers can change confidential records.

Availability

Employee accounts may be locked or misused.

- **Preventive Measures**

- Strong password policy
- Password hashing using bcrypt or Argon2
- Multi-Factor Authentication
- Password manager
- Account lockout policy

Question 2 – Security Services and Security Mechanisms

Problem Statement

A healthcare organization is implementing an Electronic Health Record (EHR) system and requires appropriate security services and mechanisms.

Requirement Analysis

Requirement 1

Only authorized doctors should access patient records.

Security Service

Authentication

Access Control

Recommended Security Mechanisms

- Multi-Factor Authentication
- Role-Based Access Control (RBAC)
- Biometric Authentication

Justification

Only verified doctors can access confidential patient records.

Requirement 2

Patient records should not be modified without authorization.

Security Service

Integrity

Mechanisms

- SHA-256 Hashing
- Digital Signatures

Justification

Hash values immediately reveal unauthorized modifications.

Requirement 3

Every medical transaction should be traceable.

Security Service

Non-Repudiation

Mechanisms

- Digital Signatures
- Audit Logs

Justification

Doctors cannot deny performing medical operations.

Requirement 4

Patient data transmitted between hospitals should remain confidential.

Security Service

Confidentiality

Mechanisms

- AES-256 Encryption
- TLS/SSL
- VPN

Justification

Encrypted communication prevents eavesdropping.

Requirement 5

System should remain operational during cyberattacks.

Security Service

Availability

Mechanisms

- Firewall

- IDS
- IPS
- Backup Servers
- DDoS Protection

Justification

Ensures continuous healthcare services.

Question 3 – Caesar Cipher

Given Ciphertext

YMNX NX F XJHWJY RJXXFLJ

Step 1: Determining the Encryption Key

Using brute-force analysis, the correct shift value is **5**.

Encryption Key = **5**

Step 2: Decryption

Y → T

M → H

N → I

X → S

The complete decrypted message becomes:

THIS IS A SECRET MESSAGE

Step 3: Why Caesar Cipher is Vulnerable

- Only 25 possible keys.
- Easily broken using brute-force attacks.
- Vulnerable to frequency analysis.
- Uses monoalphabetic substitution.
- Not secure for modern communication.

Step 4: Recommended Encryption Technique

Advanced Encryption Standard (AES-256)

Advantages

- 256-bit key
- Extremely difficult to brute force
- International encryption standard
- Used in banking, military, and cloud computing

Question 4 – PlayFier Cipher

Keyword = DEFENCE

D	E	F	N	S
A	B	C	G	H
I	K	L	M	O
P	Q	R	T	U
V	W	X	Y	Z

Step 2: Ciphertext

BMODZBXDNABEKUDMUIXMMOUIVIF

Decrypted Plaintext

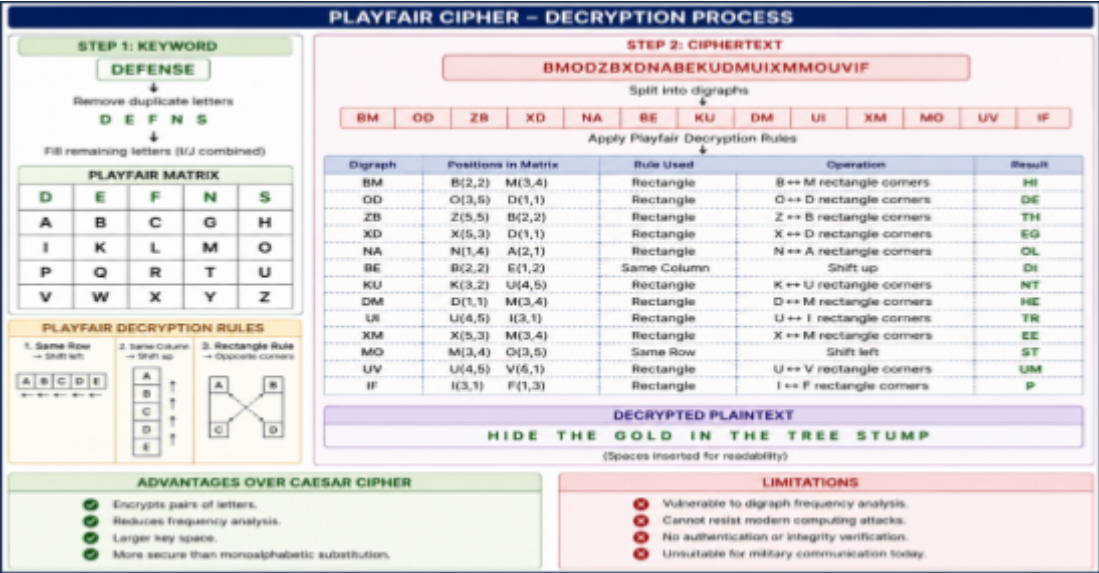
HIDE THE GOLD IN THE TREE STUMP

Advantages over Caesar Cipher

- Encrypts pairs of letters.
- Reduces frequency analysis.
- Larger key space.
- More secure than monoalphabetic substitution.

Limitations

- Vulnerable to digraph frequency analysis.
- Cannot resist modern computing attacks.
- No authentication or integrity verification.
- Unsuitable for military communication today.



Question 5 – Integrated Case Study

Security Vulnerabilities

Identified Issue

Vulnerability

Caesar Cipher

Weak Encryption

Playfair Cipher

Weak Encryption

Phishing Emails

Social Engineering

Production Schedule Modification

Weak Access Control

DDoS

Poor Network Protection

Security Attacks

Issue

Attack

Caesar Cipher

Weak Cryptography

Issue	Attack
Playfair Cipher	Weak Cryptography
Phishing	Phishing Attack
Schedule Modification	Unauthorized Access
Server Crash	Distributed Denial-of-Service

Required Security Services

Issue	Required Service
Caesar Cipher	Confidentiality
Playfair Cipher	Confidentiality
Phishing	Authentication
Unauthorized Modification	Integrity & Access Control
DDoS	Availability

Recommended Security Mechanisms

Requirement	Mechanism
Secure Communication	AES-256
Key Exchange	RSA/ECC
Authentication	Multi-Factor Authentication
Integrity	SHA-256 + Digital Signature
Access Control	RBAC
Network Security	Firewall
Threat Detection	IDS/IPS
Availability	Load Balancer + DDoS Protection
Monitoring	SIEM
Secure Transmission	TLS 1.3

Evaluation of Classical Encryption

Caesar Cipher

Not suitable because:

- Small key space
- Easily brute-forced
- Vulnerable to frequency analysis
- No authentication

Playfair Cipher

Not suitable because:

- Vulnerable to cryptanalysis
- Limited key space
- No integrity verification
- Cannot withstand modern attacks

